BioIntelliSense

16 June 2025

Stephanie Carlton
Deputy Administrator
Centers for Medicare & Medicaid Services
US Department of Health and Human Services
200 Independence Avenue, SW
Washington, DC 20201

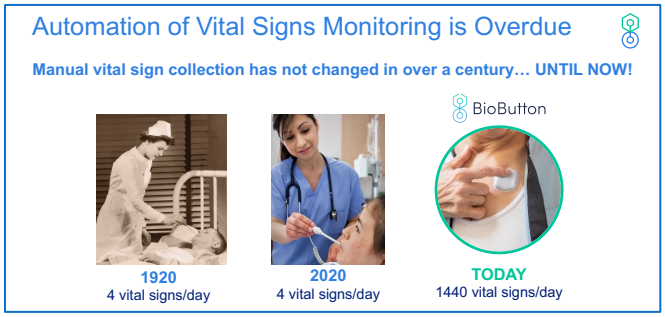
Thomas Keane
Assistant Secretary for Technology Policy
National Coordinator for Health IT
US Department of Health and Human Services
200 Independence Avenue, SW
Washington, DC 20201

RE: Health Technology Ecosystem Comments to RFI:

**In 2025, A Scalable AI-Based Wearable System for Continuous Patient Monitoring Can Dramatically Reduce Total Cost of Care While Significantly Improving Patient Outcomes**

Dear Ms. Carlton and Mr. Keane,

The monitoring of patients in the hospital (and at home) has not changed over the past 100 years…




The current state of hospital care relies on manual, intermittent snapshots of vital sign information once every 4 - 6 hours by a healthcare workforce that is challenged by staffing shortages, which jeopardizes patient safety, and drives up the cost of care. In the home setting, family caregivers are unreasonably burdened with the responsibility and cost of caring for loved ones yet have little means of recognizing the evolving complications associated with post-acute and/or chronic conditions.

In both settings, patient deterioration is commonly unrecognized until the patient "crashes," triggering the rapid response team (RRT) in the hospital, or a call to 911 and an ambulance trip to the emergency room from home. The inevitable consequences of this antiquated practice include transfers to the ICU, increased hospital mortality, increased hospital length of stay (LOS), increased healthcare workforce stress, burn-out, and labor expense, increased emergency room utilization, increased hospital readmissions, patient and family dissatisfaction, and skyrocketing increases in the total cost of healthcare to the US taxpayer and society.

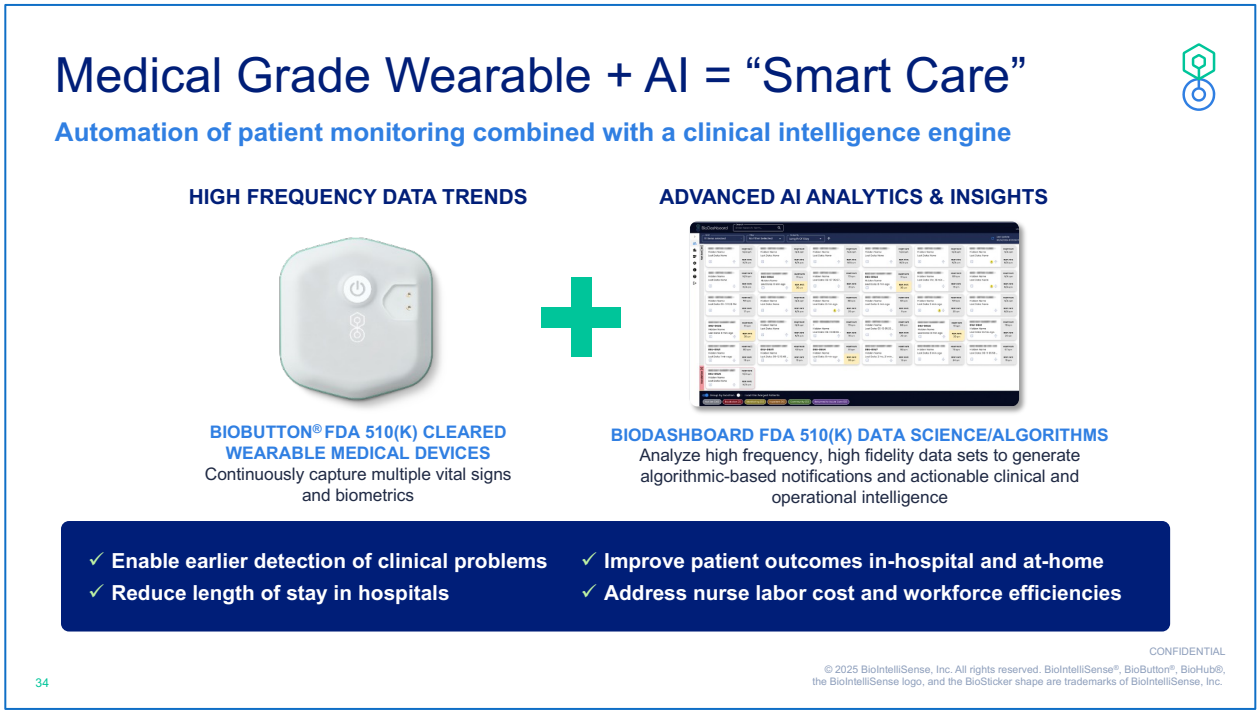
***A medical grade wearable for automated continuous patient monitoring is available today that solves each of these problems!***

**BioButton Continuous Patient Monitoring System (see attached and video link below)**
Houston Methodist BioButton WebMD Video

The BioButton Continuous Patient Monitoring System provides timely, algorithmic-based alerts as an early warning of meaningful changes in a patient's vital signs to save valuable staff time, allow for proactive clinical interventions and reduce the burden of manual charting. The BioDashboard clinical triage and intelligence system enables clinicians to receive near real-time trend analysis to help identify potential deterioration at an earlier time-frame. This system provides a comprehensive and scalable end-to-end remote care solution, to support rapid deployments across the healthcare continuum, from in-hospital to at-home.

- **FDA-cleared wearable technology:** The BioButton multi-parameter wearable device combines an effortless 'stick it on and forget it' user experience with medical grade clinical accuracy that healthcare professionals can trust. It passively captures high frequency vital signs, physiologic biometrics and symptomatic event trends on a discreet single-use, rechargeable device. The BioButton is configurable with acute and post-acute modes to address patient monitoring needs that span both care settings and acuity levels.

- **Advanced analytics and data services:** Continuous medical grade monitoring with the BioButton device automatically captures high frequency vital sign measurements, powering the BioCloud advanced analytics engine for earlier detection of subtle changes in a patients' physiologic trending status. This passive remote capture of medical grade data allows health systems to scale remote care making the home an extension of the healthcare system, while automating vital sign collection at the point-of-care, to improve clinical workflow efficiency, prioritization of care and resource allocation.

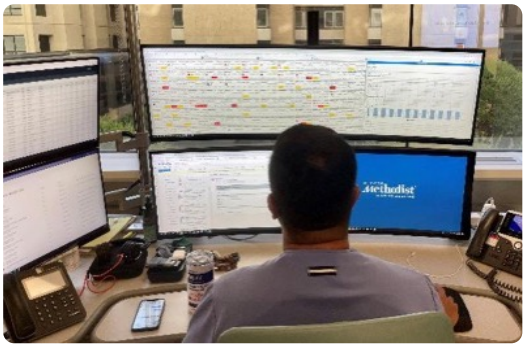


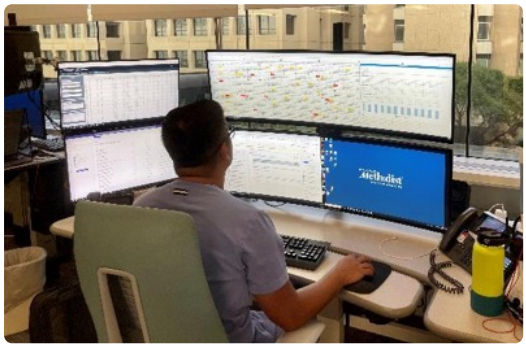
- **Data interoperatbility with modern Application Program Interfaces (APIs)** and data translation services ease the integration effort with 3rd party EMRs and enterprise systems. BioIntelliSense is agnostic to downstream systems and has built a best-in-class ecosystem with pre-integrated EMR, RPM, Telehealth, Virtual Care and Clinical Trial applications. Notably, BioIntelliSense has integrated with EPIC and leading electronic medical record systems (EMRs), utilized in the acute and post-acute settings, to support clinical workflow.

- **Clinical intelligence and triage:** BioCloud advanced analytics and algorithmic-based alerting may identify patients deemed at risk, based on vital sign trends that result in statistically meaningful change, relative to an individual's physiologic baseline. Contextual monitoring of the BioButton's high-frequency temperature, respiratory rate and heart rate at rest measurements may enable the early detection of signs and symptoms of patient deterioration. The passive collection of this continuous multi-parameter data in general care wards of the hospital and at-home, may provide better and more timely recognition of adverse trends for proactive clinical intervention and health outcomes. The benefits of this continuous care model extend beyond the hospital to the home by providing a scalable platform monitoring vital signs, physiologic biometrics and symptoms for earlier detection of adverse trends without the cost and complexity of traditional RPM.

# Continuous Patient Monitoring Command Center

## Optimize clinical resources and better prioritize care





**400 BioButton patients** per monitoring tech
and **3 monitoring techs** (1200 patients) per nurse supervisor

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**Results from full scale implementations of over 350,000 patients both in-hospital and post discharge at home:**

- In-hospital
  - Significant reductions in med/surg transfers to ICU
  - Significant reductions in APACHE severity of illness scores upon transfer to ICU
  - Significant reductions in traditional telemetry utilization
  - Significant reductions in associated ICU mortality
  - Significant reductions hospital mortality
  - Significant reductions average length of stay
  - Significant reductions in healthcare workforce labor related to manual vital sign collection and documentation

- Post acute hospital discharge:
  - Significant reductions in 30-day post discharge readmission
  - Significant reductions in 30-day post discharge ER utilization

- Long term care:
  - Significant reductions total cost of care over 12 months

In summary, this BioButton System for continuous patient monitoring is a vivid example of how technology-enabled care can significantly improve patient outcomes while dramatically reducing total cost of care. If this system were to be installed in every hospital in the US today, it would net save multiple tens of $billions per year to the Medicare and Medicaid programs. Furthermore, this can easily be implemented for any rural/critical access hospitals to address staffing shortages and keep beds open. Lastly, family caregivers would no longer be burdened by the overwhelming and unaffordable responsibility of keeping their loved ones out of emergency rooms and hospitals.

Detailed review of patient outcomes data, including publications and costs, are available on request.

Sincerely,

James Mault, MD, FACS
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